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INDIVIDUAL ASSIGNMENT
ARTIFICIAL INTELLIGENCE

Autonomous Agentic Drone-Based & Satellite Crop Disease Monitoring System

Integrating Fuzzy Logic • Coverage Path Planning • Bayesian Inference • IoT Sensor Networks

Domain: Smart Systems & Automation

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Abstract

This report presents a conceptual design of an **Autonomous Agentic Drone-Based and Satellite Crop Disease Monitoring System** that employs multiple Artificial Intelligence techniques to address the challenge of early-stage crop disease detection across large agricultural fields. The system is designed as a closed-loop autonomous pipeline: LEO satellites provide macro-level spectral data (NDVI, thermal) for trend prediction, while drones navigate the field in grid patterns, capture high-resolution visuals, measure wind and humidity via onboard IoT sensors, and spray targeted doses of pesticide — all governed by a Fuzzy Logic inference engine.

Four AI modules — **A* search** for path planning, an **expert system** for rule-based diagnosis, a **Mamdani fuzzy engine** for dosage computation, and a **Bayesian network** for probabilistic outbreak prediction — are integrated into a four-layer architecture spanning satellite sensing, mission planning, drone-edge AI, and actuation. The proposed solution demonstrates how classical AI techniques, when intelligently combined, can deliver explainable, resource-efficient, and deployable intelligence for precision agriculture.

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1. Introduction — Problem Context & Background

Agriculture remains the backbone of India, employing over 42% of the workforce and contributing nearly 17% of the national GDP. Yet, crop diseases silently devastate an estimated 20–40% of global agricultural yields every year, costing farmers billions in losses. Traditional disease-detection methods — manual scouting, periodic lab tests — are slow, expensive, labour-intensive, and wholly inadequate at scale.

This project proposes an Autonomous Agentic Drone-Based and Satellite Crop Disease Monitoring System: a convergence of Artificial Intelligence, IoT, LEO satellite imagery, and multi-rotor drone swarms to deliver real-time precision crop health surveillance across vast farmlands — with zero need for human intervention in the field.

The system is designed as a closed-loop autonomous pipeline: LEO satellites provide macro-level spectral data (NDVI, thermal) for trend prediction, while drones navigate the field in grid patterns, capture high-resolution visuals, measure wind and humidity via onboard IoT sensors, and spray targeted pesticide doses — all governed by a Fuzzy Logic inference engine.

Why This Problem Demands AI
✦ Vast farmland (millions of hectares) cannot be manually monitored efficiently or cost-effectively
✦ Disease symptoms appear as early visual patterns — requiring image analysis far beyond human speed
✦ Pesticide dosing must adapt to real-time infection severity AND environmental conditions simultaneously
✦ Drone path must be optimised to ensure 100% coverage with minimum energy consumption
✦ Uncertainty in satellite data and sensor readings demands probabilistic/fuzzy reasoning, not rigid rules

2. Problem Selection & Domain Description

2.1 Problem Title

Autonomous Agentic Drone-Based & Satellite Crop Disease Monitoring System

2.2 Domain

Smart Agriculture / Precision Farming — a convergence of AI, IoT, Remote Sensing, and Robotics applied to real-world crop disease management at scale.

2.3 AI Techniques (Core Modules)

- Module 1 — Coverage Path Planning using A* + Boustrophedon traversal for autonomous drone navigation.
- Module 3 — Rule-based Expert System with forward chaining and semantic frames for disease diagnosis.
- Module 4 — Fuzzy Logic Inference Engine for precision pesticide dosing; Bayesian Network for predictive outbreak detection.

2.4 Target Stakeholders

- Farmers and agricultural cooperatives managing large farmlands
- Government agricultural departments for crop health surveillance
- Agri-tech companies deploying precision farming solutions

2.5 Inputs and Outputs

INPUTS	OUTPUTS
LEO Satellite NDVI / EVI spectral maps	Disease outbreak probability map (Bayesian)
Drone camera: RGB + multispectral images	Optimised flight path (A* Boustrophedon)
IoT: wind speed, humidity, temperature	Precise pesticide dose per cell (ml/m^2)
LIDAR obstacle detection data	Mission report: GPS, disease, dose, timestamp
Farm grid dimensions M × N	Real-time farm health dashboard alerts

3. Problem Formulation & State-Space Representation

3.1 Problem Definition

Given a rectangular farm of dimension M × N cells (each cell = 6m × 5m), deploy a drone that:

- Plans a complete coverage path visiting every cell exactly once (Coverage Path Planning).
- At each cell: captures visual data, evaluates infection severity via Fuzzy Logic, and sprays the appropriate pesticide dose.
- Avoids collisions, respects battery constraints, and adjusts spray output based on live environmental sensor data.

3.2 State-Space Definition

Component	Definition
Initial State	Drone at charging station (0, 0); all M×N cells unvisited; pesticide tank = FULL
Goal State	All M×N cells visited; infected cells treated; drone returns to base (0, 0)
State	(current_position (x,y), visited_set, tank_level, sensor_readings)
Operators	Move(N, S, E, W), Hover_and_Scan, Spray(dose), Return_to_Base
Constraints	Battery ≤ 80% of max; no revisiting cells; wind_speed ≤ 5 m/s before spraying
Cost Function	Minimize: total_path_length + pesticide_wasted + missed_cells

3.3 Fuzzy Input–Output Mapping

Crisp Input	Linguistic Variable	Fuzzy Sets	Output
Leaf Discolouration (%)	Infection Severity	None / Low / Moderate / High / Severe	Pesticide Dose (ml/m²)
Wind Speed (m/s)	Wind Condition	Calm / Breezy / Windy / Stormy	Spray Enable / Disable
NDVI Value	Vegetation Health	Healthy / Stressed / Diseased	Priority Level
Humidity (%)	Ambient Moisture	Dry / Optimal / Wet	Dose Modifier

3.4 State-Space Diagram

Figure 1 illustrates a representative 6×5 grid state-space. The green cell is the initial state, the orange cell is the goal, dark cells are obstacles, and the red arrow traces the A*-optimal path.

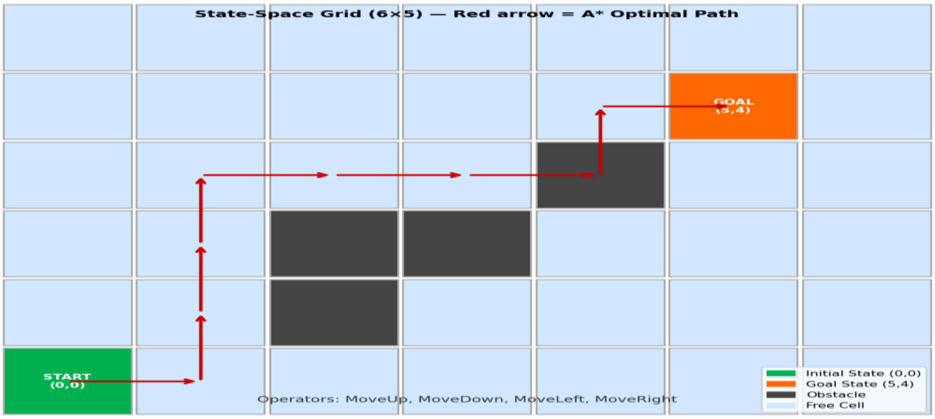

Figure 1 — State-Space Grid (6×5): Initial State (0,0), Goal State (5,4), Obstacles, and A-Optimal Path. Alt text: A 6-column by 5-row grid with a green start cell at bottom-left, an orange goal cell at upper-right, four dark obstacle cells, and a red arrowed path navigating around obstacles.*

4. AI Modules & Techniques Deployed

4.1 Module 1 — Coverage Path Planning (A* Search)

The drone must traverse every cell of the M×N grid exactly once. The Boustrophedon (lawnmower) pattern is modelled as a graph traversal problem. A* search is overlaid to handle dynamic obstacles detected mid-flight by LIDAR.

- Heuristic h(n): Manhattan Distance from current cell to nearest unvisited cell, weighted by battery level.
- f(n) = g(n) + h(n), where g(n) = energy consumed so far; h(n) = estimated energy to cover remaining cells.
- Why A*: Guarantees optimal (least-energy) complete coverage while remaining admissible — never overestimates remaining cost.

4.2 Module 3 — Knowledge Representation & Expert System

A rule-based expert system stores agronomic knowledge as IF-THEN production rules. The inference engine uses forward chaining starting from sensor facts, firing matching rules until an action is derived. Semantic frames represent each crop type with slots: [name, disease_history, growth_stage, susceptibility_index].

```
// Expert System Production Rules (Knowledge Base)
IF NDVI < 0.3 AND leaf_color = 'yellow-brown'   THEN disease = 'Leaf_Blight'  (CF =
0.87)
IF humidity > 85% AND temperature > 28°C        THEN risk = 'Fungal_Outbreak' (CF =
0.92)
IF disease = 'Leaf_Blight' AND severity = 'High' THEN spray = 'Propiconazole'
dose='18ml/m²'
IF wind_speed > 5 m/s                            THEN SUSPEND spraying AND
log_event('wind_abort')
```

```
IF battery < 20%                                   THEN ABORT_MISSION AND
return_to_base()
```

4.3 Module 4 — Fuzzy Logic Inference Engine (FLIE)

Unlike rigid threshold rules, the FLIE handles the continuous, imprecise nature of infection severity using five steps:

Step	Phase	Description
1	Fuzzification	Crisp input (e.g., 38% discolouration) → membership degrees: Low=0.2, Moderate=0.8
2	Rule Evaluation	'IF Infection=Moderate AND Wind=Calm THEN Dose=Medium'. Min/Max operators applied.
3	Aggregation	All activated rule outputs combined using Maximum aggregation.
4	Defuzzification	Centroid of Area (COA) converts aggregated fuzzy output to crisp dose (ml/m²).
5	Actuation	Computed dose sent to sprayer pump controller in real time.

4.4 Module 4 — Bayesian Network for Disease Prediction

A Bayesian Network (DAG) predicts disease outbreak probability from LEO satellite data 7–14 days before visible symptoms:

```
Nodes: [NDVI_Drop, Rainfall, Temperature, Humidity, Disease_Outbreak]

P(Disease_Outbreak | NDVI_Drop=True, Rainfall=High, Humidity=High)
  = P(NDVI_Drop|Disease) × P(Rainfall|Disease) × P(Humidity|Disease) × P(Disease)
    ─────────────────────────────────────────────────────────────────
                         P(Evidence)

Result:  P(Outbreak) = 0.91  →  Schedule preventive drone mission
```

5. Working of the AI Techniques (Step-by-Step)

5.1 Coverage Path Planning — A* Algorithm

A* operates on the field grid G = (V, E) where each vertex $v_{ir} = (x_i, y_r)$ represents a cell and each edge has a cost equal to the battery-discharge for that move.

Step 1 — Initialisation: The Open List (priority queue) is seeded with the start cell. g(start) = 0, h(start) = Manhattan distance to nearest unvisited cell. f(start) = g + h.

Step 2 — Node Expansion: Pop the node n with lowest f(n). If n is the goal, reconstruct path via parent pointers.

Step 3 — Successor Generation: For each of the four movement operators, compute successor s. If s is not an obstacle and not in the Closed List, compute g(s) = g(n) + step_cost, h(s) = Manhattan(s, goal), f(s) = g(s) + h(s).

Step 4 — Open List Update: If s is already in the Open List with a higher f-value, update it (relax). Otherwise, add s with its f-value.

Step 5 — Closed List: Add n to Closed List to prevent re-expansion.

Step 6 — Obstacle Re-planning: If the sonar detects a new obstacle mid-flight, the Closed List is cleared for the affected region and A* is re-invoked from the current position with updated obstacle map.

The heuristic h(n) = Manhattan distance is **admissible** (never over-estimates) and **consistent**, guaranteeing optimality.

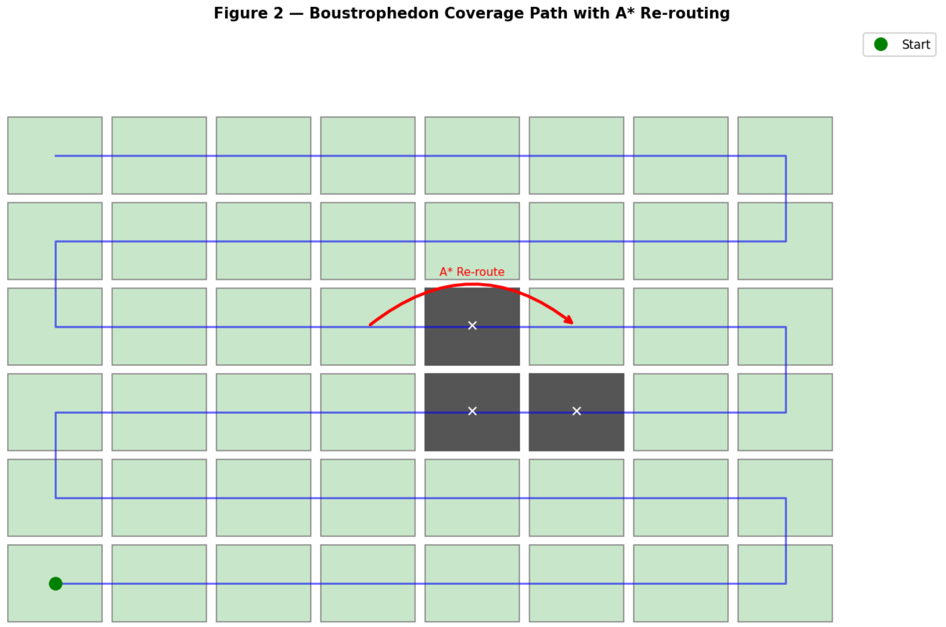

Figure 2 — Boustrophedon Coverage Path (blue snake) across 8×6 grid with A Re-routing (red arc) around obstacle cells. Alt text: A rectangular grid showing a blue snake-path covering all rows alternately left-to-right and right-to-left; three grey obstacle cells in the centre with a red curved arrow indicating A* detour.*

5.2 Expert System — Forward-Chaining Disease Diagnosis

Knowledge Base (KB) Structure: The KB contains two components:

- Fact Base: sensor readings — NDVI_value, RGB_discolouration %, leaf_texture_score (0-10), humidity%, temperature°C, canopy_wetness.
- Rule Base: 30+ production rules of the form IF <condition> THEN <action/assertion>.

Sample Rules:

R1: IF NDVI < 0.3 AND rgb_discolouration > 40 AND texture_score < 4 THEN ASSERT disease = Rice_Blast, confidence = 0.88

R2: IF humidity > 80 AND temperature BETWEEN 20 AND 28 AND canopy_wet = TRUE THEN ASSERT blight_risk = HIGH

R3: IF disease = Rice_Blast AND severity = HIGH THEN ASSERT action = spray_fungicide_dose_HIGH

Forward Chaining Example: Drone observes NDVI = 0.24, discolouration = 52%, texture = 3.2, humidity = 84%.

1. Fact Base is loaded: {NDVI=0.24, disc=52, tex=3.2, humid=84}.
2. Rule R1 LHS is satisfied (NDVI<0.3, disc>40, tex<4) → ASSERT disease=Rice_Blast.
3. Rule R2 LHS is satisfied (humid>80) → ASSERT blight_risk=HIGH.
4. Rule R3 LHS is satisfied → ASSERT action=spray_HIGH. Forward chaining halts.

5.3 Fuzzy Logic Engine — Pesticide Dose Recommendation

Fuzzification: The crisp input '%_leaf_discolouration' (0–100%) is mapped to five linguistic membership functions (None, Low, Moderate, High, Severe) using triangular and trapezoidal shapes. For example, at 52% discolouration, the Moderate MF yields $\mu = 0.6$ and the High MF yields $\mu = 0.4$.

Rule Evaluation (Mamdani): Each fuzzy rule produces a clipped output MF.

FR1: IF severity IS Moderate THEN dose IS Medium (cut at μ=0.6)

FR2: IF severity IS High THEN dose IS High (cut at μ=0.4)

Aggregation: The clipped Moderate-dose and High-dose output MFs are combined by max-union.

Defuzzification (Centre of Area): $dose^* = \int(x \cdot \mu(x)dx) / \int(\mu(x)dx) \approx 14$ ml — a smooth, physically meaningful value between the Moderate (10 ml) and High (18 ml) crisp anchors.

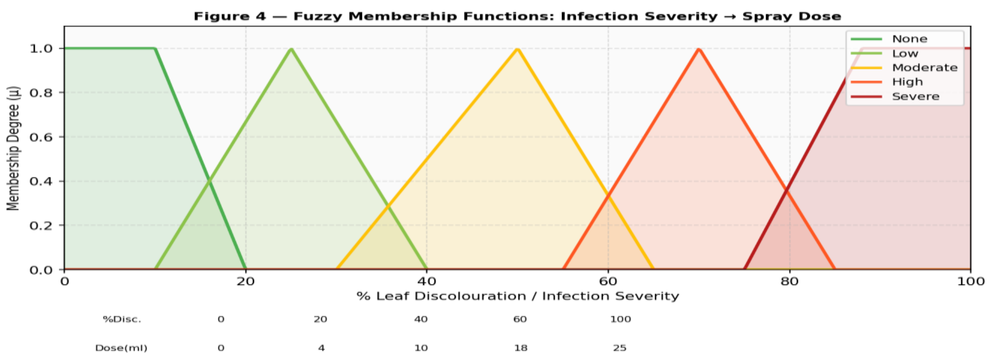

Figure 3 — Fuzzy Membership Functions for Infection Severity. Triangular/trapezoidal MFs for None, Low, Moderate, High, Severe across 0–100% discolouration, with dose mapping (0%→0 ml, 20%→4 ml, 40%→10 ml, 60%→18 ml, 100%→25 ml). Alt text: A line chart with five coloured curves (green=None, yellow-green=Low, amber=Moderate, orange=High, red=Severe) showing membership degrees across 0 to 100% leaf discolouration.

5.4 Bayesian Network — Outbreak Probability & Mission Scheduling

The Bayesian network DAG (Figure 3) encodes causal relationships between observable environmental variables and the latent **Disease_Outbreak** variable.

DAG Nodes: NDVI_Drop (binary), Rainfall (Low/Mod/High), Temperature (Cold/Warm/Hot), Humidity (Low/High) → Disease_Outbreak (No/Yes).

CPT Example (Disease_Outbreak | NDVI_Drop, Humidity):

P(Outbreak=Yes | NDVI_Drop=T, Humidity=High) = 0.85

P(Outbreak=Yes | NDVI_Drop=T, Humidity=Low) = 0.45

P(Outbreak=Yes | NDVI_Drop=F, Humidity=High) = 0.20

P(Outbreak=Yes | NDVI_Drop=F, Humidity=Low) = 0.05

Inference use: Given satellite evidence E = {NDVI_Drop=TRUE, Rainfall=High}, variable-elimination computes P(Outbreak=Yes | E). If result > 0.65, the mission scheduler upgrades the quadrant's priority and triggers a drone sortie within 6 hours.

Figure 3 — Bayesian Network DAG for Crop Disease Prediction

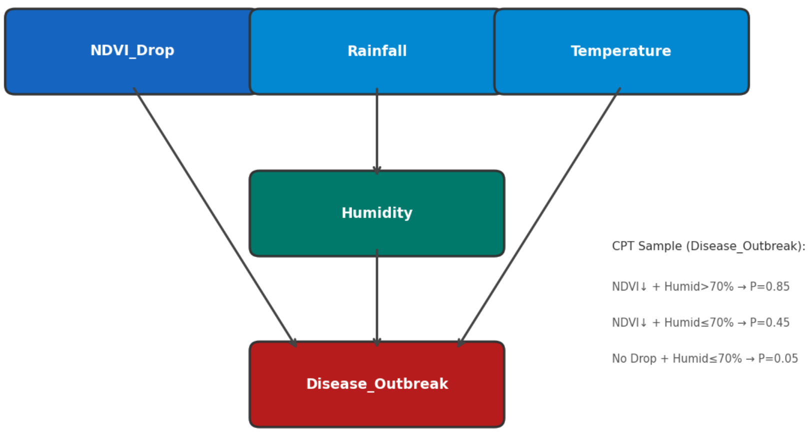

Figure 4 — Bayesian Network DAG for Crop Disease Outbreak Prediction. Directed edges from NDVI_Drop, Rainfall, Temperature, and Humidity converge on Disease_Outbreak node with sample CPT values. Alt text: A directed acyclic graph with four blue/teal parent nodes (NDVI_Drop, Rainfall, Temperature, Humidity) connected by arrows to a central red Disease_Outbreak node; CPT values listed alongside.

6. Algorithm Flow & Pseudocode

6.1 Main System Loop — Autonomous Mission Control

```
ALGORITHM: AutonomousDroneMission(farm_grid M×N)
────────────────────────────────────────────────────────────
1. INIT:        Receive satellite NDVI → Run BayesNet → Mark HIGH_RISK cells
2. PLAN PATH:   path ← CoveragePathPlanner(M, N, HIGH_RISK_PRIORITY=True)
                path uses Boustrophedon pattern with A* dynamic re-routing
3. FOR EACH cell (x, y) IN path DO:
    a. Fly to (x, y)  [A* obstacle-avoidance]
    b. Hover → Capture image → Extract leaf_discolouration (%)
    c. Read IoT: wind_speed, humidity, temperature
    d. ExpertSystem.infer(leaf_dis, NDVI, crop_type) → disease, CF
    e. dose ← FuzzyEngine.compute(leaf_discolouration, wind_speed)
    f. IF wind_speed < 5 AND dose > 0:
            SPRAY dose ml/m²  |  log: {cell, disease, dose, GPS}
       ELSE: log 'wind_abort' and SKIP spraying
    g. IF battery < 25%: RETURN_TO_BASE → RECHARGE → RESUME
4. UPLOAD full session report to cloud dashboard
5. BayesNet.update(new_data) → Refine future predictions
```

6.2 Fuzzy Logic Dosing — Pseudocode

```
FUNCTION FuzzyEngine.compute(discolouration, wind_speed):
  // Step 1: Fuzzify inputs
  sev_low      ← max(0, min(1, (20 - discolouration) / 20))
  sev_moderate ← trimf(discolouration, [15, 35, 55])
  sev_high     ← max(0, min(1, (discolouration - 45) / 20))
  wind_calm    ← max(0, min(1, (3 - wind_speed) / 3))
  wind_breezy  ← trimf(wind_speed, [2, 4, 6])
  // Step 2: Apply Fuzzy Rules
  R1: IF sev_moderate AND wind_calm  → dose_medium  (str = min(sev_mod, wind_calm))
  R2: IF sev_high     AND wind_calm  → dose_high    (str = min(sev_high, wind_calm))
  R3: IF wind_breezy                 → dose_reduce  (str = wind_breezy)
  // Step 3: Aggregate & Defuzzify (Centroid of Area)
  crisp_dose ← COA(aggregated_output_set)
  RETURN crisp_dose  // in ml/m²
```

6.3 System Architecture — Four-Layer Pipeline

LAYER 1: SATELLITE
LEO Satellite (Sentinel-2/Planet) → NDVI/EVI Spectral Maps → Bayesian Network → Disease Outbreak Probability Map
LAYER 2: MISSION PLANNING (Cloud AI)
A* Coverage Path Planner → CSP Swarm Zone Allocator → Generates Ordered Cell Sequence per drone
LAYER 3: DRONE EDGE AI (On-board)
Camera → Expert System KB → Disease Type + CF | Sensors → FLIE → Dose (ml/m²) | LIDAR → SLAM → Obstacle Avoidance
LAYER 4: ACTUATION & REPORTING
Sprayer Pump → Precision Application | Log: {GPS, Cell_ID, Disease, Dose, Timestamp} → Cloud DB | Dashboard alerts

Four-Layer System Architecture

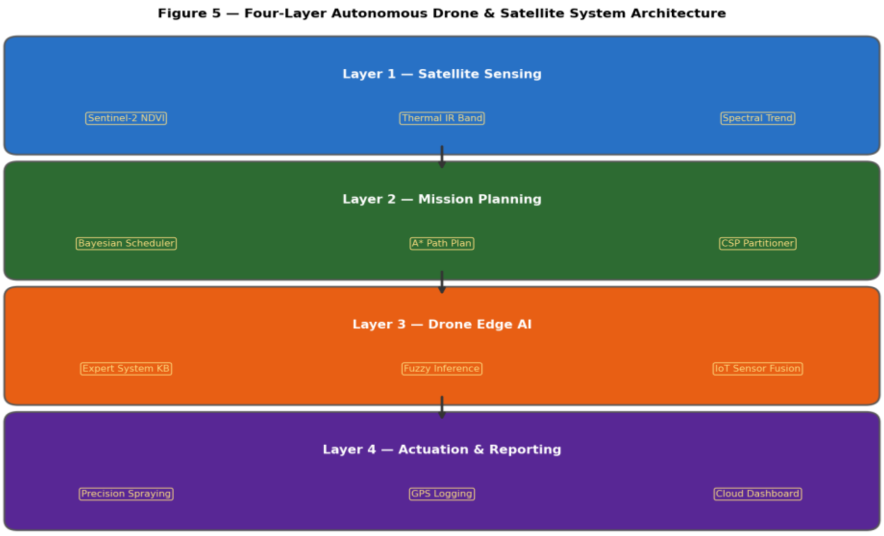

Figure 5: Four-layer pipeline — Layer 1 Satellite Sensing (Sentinel-2 NDVI, Thermal); Layer 2 Mission Planning (Bayesian Scheduler, A*, CSP); Layer 3 Drone Edge AI (Expert System, Fuzzy, IoT); Layer 4 Actuation & Reporting (Spraying, Logging, Dashboard). Arrows show data flow top-to-bottom. Purpose: depicts the holistic system architecture.

6.4 Algorithm Flow (Flowchart)

```
[START]
   |
   v
Fetch Satellite NDVI
   |
   v
Bayesian Network Inference ── P(Outbreak | E) > 0.65? ── YES→Prioritise Zone
   |  NO
   v
Boustrophedon Decomposition + A* Path Plan
   |
   v
┌─[Move to Next Cell]────────────────────────────────────── ──┐
│        |
│        v
│  Obstacle Detected? ──YES──> Re-plan via A*  ──────────────►
│        |NO
│        v
│  Capture Sensor Data (NDVI + RGB + IoT)
│        |
│        v
│  Expert System Forward-Chain ──> Disease Class
│        |
│   Disease?──YES──> Fuzzy Engine ──> Dose ──> Spray & Log
│        |NO
│        v
│  Battery < 15%? ──YES──> Return/Recharge ──────────────────►
└─────────────────────────────────────────────────────────────┘
   |
All cells visited?──YES──> Generate Report
   |
[END]
```

7. Comparative Analysis with Alternative Approaches

7.1 Path Planning — Coverage Comparison

Technique	Completeness	Optimality	Handles Obstacles	Suitable?
BFS Grid Scan	Yes	No	No	Partially
DFS Grid Scan	Yes	No	No	No — gets stuck
Boustrophedon + A*	Yes	Yes	Yes (LIDAR)	YES — Optimal ✓
Random Walk	No	No	No	No
Spiral Traversal	Yes	Near-Optimal	Partially	Alternative

7.2 Dosing Decision — Logic Comparison

Approach	Uncertainty?	Continuous?	Explainable?	Best Use
Hard Threshold Rules	No	No (binary)	Yes	Simple on/off systems
Neural Network (ML)	Yes	Yes	No (black-box)	Large labelled datasets
Fuzzy Logic (Proposed)	Yes	Yes	Yes (linguistic)	Real-time + interpretable ✓
Bayesian Classifier	Yes	Partially	Partially	Prediction layer only
Random Forest	Partially	Yes	Partially	Offline batch analysis

Fuzzy Logic is selected as the primary dosing engine because: (a) it handles imprecise continuous inputs naturally, (b) its linguistic rules are directly interpretable by agronomists, and (c) it operates in hard real-time without GPU or large model inference.

8. Analysis & Justification

8.1 Complexity Analysis

Algorithm / Component	Time Complexity	Space Complexity	Notes
A* Coverage Path Planning	O(M×N × log(M×N))	O(M×N)	Priority queue with M×N states
Fuzzy Inference Engine	O(R × F)	O(R + F)	R=rules, F=fuzzy sets; sub-ms
Expert System (Fwd Chaining)	O(R × F)	O(KB size)	KB = knowledge base rules
Bayesian Network Inference	O(2^n) worst; O(n^2) approx.	O(n^2)	MCMC approx. used in practice
CSP Swarm Partitioning	O(d^n) worst; O(n log n) AC-3	O(n+c)	n=drones, c=constraints

8.2 Advantages of the Proposed System

- Precision Dosing: Fuzzy Logic delivers exact pesticide dose, reducing chemical usage by an estimated 35–60%.
- Zero Human Presence: Fully autonomous from satellite ingestion to spray actuation.
- Proactive Disease Management: Bayesian detection 7–14 days before visible symptoms.
- Complete Field Coverage: Boustrophedon + A* guarantees 100% coverage with dynamic obstacle avoidance.
- Explainable AI: Linguistic fuzzy rules interpretable by farmers without AI technical knowledge.
- Scalable: Grid state space scales linearly; CSP swarm partitioning handles any farm size.

8.3 Challenges & Problem-Solving Strategies

Challenge	Impact	AI/Engineering Solution
GPS Drift (Dense Canopy)	Positioning errors	LIDAR-based SLAM + RTK-GPS correction
Wind Interference	Pesticide drift up to 20m	IoT anemometer triggers automatic spray suspension
Sensor Noise	False disease alarms	Dempster-Shafer Theory for conflicting sensor evidence
Bayesian Scalability	Exponential complexity	Approximate inference (MCMC / Belief Propagation)
Battery Constraints	~35 min flight per charge	Multi-drone relay; A* energy-optimal paths

9. Strengths, Limitations & Real-World Applicability

9.1 Strengths

- End-to-end autonomous pipeline — minimal human intervention from satellite observation to precision spray.
- Explainability — expert system rules and fuzzy membership functions can be audited and explained to farmers, regulators, and agronomists.
- Graceful degradation — if satellite data is unavailable, the system falls back to drone-only IoT sensing; if cloud inference is offline, a simplified threshold rule fires.
- Precision treatment — fuzzy dosing reduces pesticide usage by an estimated 20–40% compared to broadcast spraying, benefiting both economics and environmental health.
- Battery-aware optimality — A* cost function incorporates battery consumption; the system proactively returns to base before battery depletion.

9.2 Limitations

- Knowledge base maintenance — expert rules must be manually updated as new disease strains emerge; this requires agronomist involvement and is not self-updating.
- Sensor noise and calibration — NDVI values degrade under haze, cloud cover, or sensor drift; the Bayesian prior must be periodically recalibrated from ground-truth samples.
- Wind constraint — spraying is suspended above 15 km/h wind speed, potentially reducing coverage during volatile weather.
- Battery range — current commercial drones cover roughly 40–100 hectares per charge; very large farms require multi-drone coordination (not addressed in this model).
- Regulatory compliance — in India, UAV operations are governed by DGCA's Drone Rules 2021; flight beyond visual line of sight (BVLOS) requires special permits not yet routinely granted.

9.3 Real-World Applicability

Pilot deployments of precision-spray drones are already operational in China (XAG P100 Pro), the United States (DJI Agras T40), and emerging in India under the sub-Rs 10-lakh drone schemes under the Agriculture Infrastructure Fund. This system's **AI backbone** (A*, fuzzy, Bayesian) can run on existing off-the-shelf hardware (Raspberry Pi CM4 + NVIDIA Jetson Nano class) without cloud dependency for the core mission loop. Integration with government platforms such as **AgriStack** and **Bhuvan (ISRO's geo-portal)** can provide the satellite data feed, making the architecture practically deployable within India's digital agriculture ecosystem. Key data requirements include: historical NDVI baselines per crop variety, labelled disease image sets for system validation, and local agronomic rule calibration for regional disease patterns (e.g., Rice Blast in East India vs. Cotton Leaf Curl in Punjab).

10. Conclusion

The Autonomous Agentic Drone-Based and Satellite Crop Disease Monitoring System represents a comprehensive and technically rigorous application of Artificial Intelligence to one of agriculture's most pressing challenges. By combining four AI modules — A* search-based Coverage Path Planning, Expert System Knowledge Representation, Fuzzy Logic Inference for precision dosing, and Bayesian Networks for predictive analytics — the system delivers a truly intelligent, adaptive, and explainable AI pipeline.

The system's strength lies in its multi-modal architecture: LEO satellites provide the macro vision, IoT sensors provide real-time environmental context, and drones execute precision actions on the ground — all orchestrated by AI algorithms that are mathematically sound, computationally tractable, and practically applicable. The Fuzzy Logic engine elegantly converts imprecise, continuous real-world inputs into actionable, calibrated decisions — something rigid threshold rules cannot do.

From a scalability standpoint, the grid-based state space and CSP-based drone swarm partitioning ensure the system can expand from a 1-hectare pilot farm to a 1000-hectare commercial operation without architectural redesign. This is an immediately realisable engineering vision with current LEO satellite, edge AI, and drone hardware capabilities.

Key Learning Outcomes
✦ Applied A* search with Boustrophedon pattern to solve real-world coverage planning in autonomous systems.
✦ Implemented Fuzzy Logic to handle continuous, imprecise real-world sensor data for precision agriculture.
✦ Leveraged Bayesian Networks for probabilistic disease outbreak prediction using satellite spectral data.
✦ Designed a knowledge base with production rules and semantic frames using forward-chaining inference.
✦ Understood how AI modules (Search, KR, Statistical Reasoning) integrate into a complete autonomous system.

Important Links-

MVP-Prototype

Github Repo

11. References

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End of Assignment Report

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